International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

CIP-B101

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

Submitted by

JOY EJIDE

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Platform Used: Windows 11 Pro

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Introduction

The purpose of this laboratory exercise was to examine the fundamental components of a computer system from a digital forensic perspective. The lab involved identifying system hardware and software specifications, examining storage devices and partitions, investigating running processes, calculating partition sizes, understanding the boot process, and identifying file systems. All observations were performed using read-only forensic techniques to preserve system integrity and maintain evidential reliability.

Item	Value
OS Name	Microsoft Windows 11 Pro
Version	10.0.26300 Build 26300
System Manufacturer	HP
System Model	HP EliteBook 840 14-inch G10 Notebook PC
System Type	x64-based PC
Processor	13th Gen Intel Core i5-1345U
Bios Mode	UEFI
Secure Boot state	On

Observation

The machine is a modern Windows 11 laptop operating in UEFI mode with Secure Boot enabled. This configuration supports GPT partitioning and enhanced startup security.

Storage Information

Physical Disk

Item	Value
Disk Number	Disk 0

Storage Device	SK Hynix PC801 NVMe SSD
Total Capacity	476.94 GB
Partition Style	GPT
Health Status	Healthy

Observation

The storage device is an NVMe solid-state drive using GPT partitioning, which is the standard partitioning format for UEFI-based Windows systems.

Layer / Source	**Example Evidence**	**Location**	**Why important**
User layer	Searches, usernames, opened documents	Browser history, recent files	Links system activity to a user
Application layer	Office metadata, browser cache, chat logs	Application databases and folders	Shows which software created evidence
Operating system layer	Event logs, installed software, user accounts	Registry, Event Viewer	Records system activity
Disk layer	Partitions, deleted files, file fragments	Registry, Event Viewer	Supports evidence recovery
Memory layer	Running processes	RAM	Captures live system

Why Evidence Exists in Multiple Layers

A single user activity often creates evidence in multiple locations. For example, opening a document may generate browser records, application metadata, operating system logs, memory artefacts, and disk entries. Correlating evidence from multiple layers strengthens forensic findings.

Part B: Program, Process and Thread Investigation

Sample Processes Identified

Process Name	PID	Purpose
System	4	Core Windows kernel process
winlogon.exe	1284	User logon management
lsass.exe	1364	Authentication and security management

Program, Process and Thread

A **program** is a file stored on disk containing executable instructions.

A **process** is a running instance of a program loaded into memory.

A **thread** is the smallest unit of execution within a process.

Forensic Importance

Process information assists investigators in:

- Detecting malware activity
- Identifying active programs
- Determining user actions
- Examining memory-resident threats
- Performing live system triage

Part C: Disk Sector and Disk Size Calculation

Tools Used

Tools	Purpose
System Information (msinfo32)	System specification collection
Command Prompt	System and storage information
PowerShell	Disk, partition and volume inspection

Task Manager / task list	Process investigation
Disk Part	Disk inspection
Calculator	Manual partition size calculations
Snipping Tool	Screenshot collection

Part E: Boot Process Explanation

Boot Sequence

1. Power button is pressed.
2. POST performs hardware checks.
3. UEFI firmware initializes hardware.
4. Firmware loads configuration settings.
5. GPT partition information is read.
6. EFI System Partition is located.
7. Windows Boot Manager starts.
8. Windows kernel loads into memory.
9. Device drivers initialize.
10. Windows gains control and presents the login screen

Forensic Relevance of Boot Stages

Stage	Forensic Relevance
POST	Confirms hardware initialization
UEFI	Stores firmware settings
GPT	Contains partition information
Boot Manager	Controls operating system startup
Kernel Loading	Begins operating system logging

Evidence of Boot Mode

The examined system operates in:

Plain Text

BIOS Mode: UEFI

Secure Boot: Enabled

Evidence Layer Mapping

Why Evidence Exists in Multiple Layers

A single action performed by the user can leave artefacts in several locations. For example, opening a document may generate browser records, application metadata, event logs, memory artefacts, and changes within the file system. Examining multiple evidence sources improves the reliability of forensic findings.

Process Observation

Conclusion

This laboratory exercise provided practical experience in system identification, storage examination, process analysis, partition inspection, and file system analysis using forensic methods. The investigation confirmed that the examined computer uses a GPT-partitioned SK hynix NVMe SSD, UEFI firmware, Secure Boot, and NTFS file systems. Partition sizes were manually verified and matched the values reported by the operating system. The lab demonstrated the importance of collecting and validating evidence using repeatable, non-destructive methods, which is a fundamental principle of digital forensic investigations.